IPS ACADEMY, INDORE · CSE–AIML · B.TECH MAJOR PROJECT 2026–27

SENTINEL

AI-Powered Financial Crime Intelligence

Transaction Screening · Fraud Detection · Merchant Approval

"Behind every alert is a person trying to do the right thing — we built Sentinel so machines catch the noise and humans keep the trust."

SHREYASH SINGH & Team · Guide: Prof. (CSE–AIML) · RGPV University

The human cost of financial crime

Why one platform can change how institutions protect people and money.

Every day, institutions drown in a flood of alerts. Most are false — and each one steals time from an analyst who could be stopping real harm.

- **Money laundering:** **2–5% of global GDP ($0.8–2T/yr)** — FATF
- 94% of AML alerts are **false positives**, reviewed at **$50–$100 each**
- Fraud rings hide in **relationships** no single rule can see
- 311% rise in **synthetic-identity fraud** in a single year
- **UPI alone:** **15B+ transactions every month**

$20B+

spent yearly on AML compliance worldwide

94%

of alerts are false alarms

94%

miss simplest fraud patterns

Legacy rules can't see the story

context-blind, static, and siloed — until now

One platform, three protections, people first

Sentinel unifies screening, fraud detection, and merchant approval on one real-time spine.

PILLAR 1 · SCREEN

Catch bad actors

AML & sanctions checks on every transaction, with entity resolution that understands names — not just strings.

PILLAR 2 · DETECT

Stop fraud early

Behavioral profiling + ML + graph find coordinated fraud rings a single rule misses.

PILLAR 3 · APPROVE

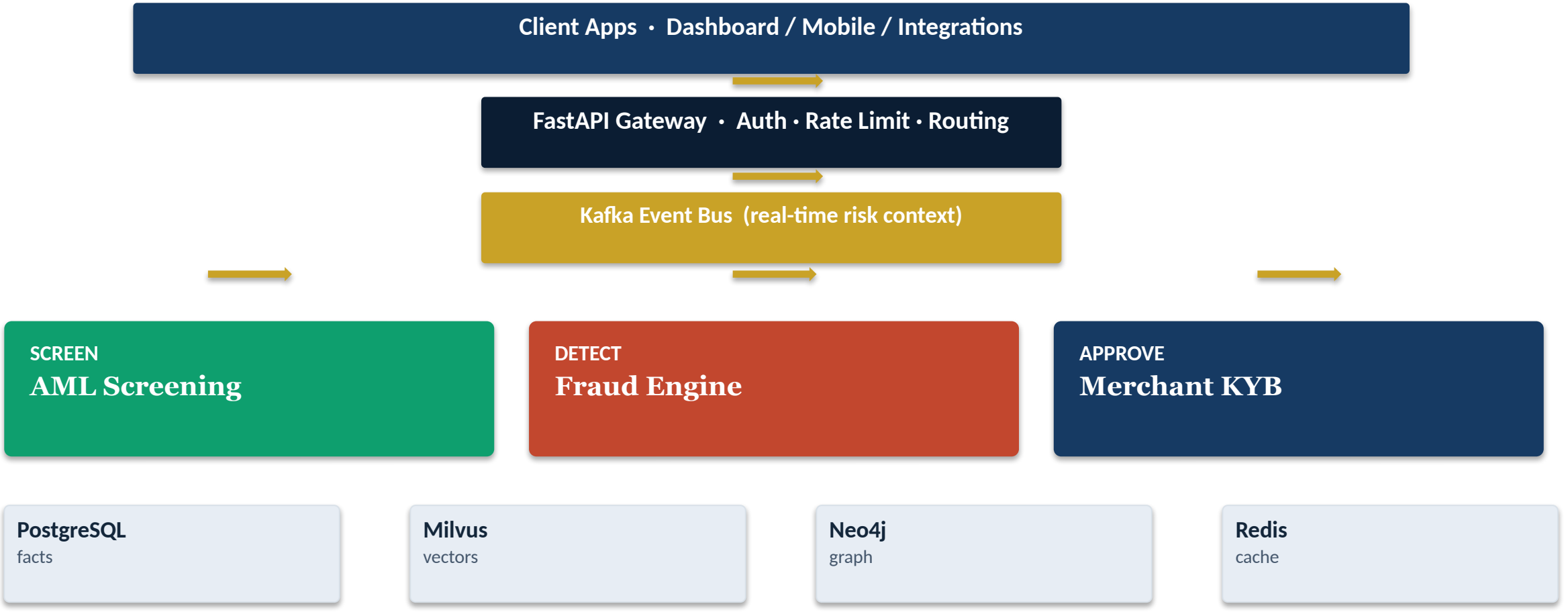
Trust good merchants

KYB/KYC in minutes, not days — rigorous, explainable, fair to honest businesses.

Why it matters to people fewer false alarms means analysts investigate what truly matters; faster onboarding gets real businesses to market; cleaner screening keeps every customer's money safer.

One event-driven spine, three risk engines

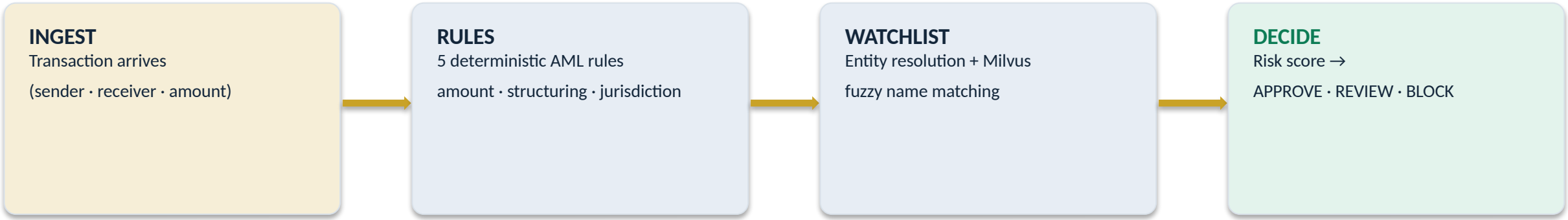
A single real-time pipeline unifies screening, fraud, and onboarding so risk context is shared everywhere.



Docker Compose deploys the full 10-service stack with one command — designed Kubernetes-ready.

Transaction Screening — AML & sanctions

Every transaction is checked against rules and watchlists before money moves.



Output: risk_score · flags · matched entities · recommendation · human-readable explanation — every decision audited

- Entity resolution understands **names, not strings** — aliases, transliterations, typos
- Vector similarity in **Milvus** catches matches exact lookup misses
- Thresholds: **REVIEW ≥ 0.40 · BLOCK ≥ 0.70**

The human layer

Analysts see a clear reason, not a raw score — so a blocked transfer can be explained to the customer who simply made a mistake.

Real-time fraud detection that learns the user

Behavioral profiling + ML + relationship graphs — then an agent that investigates before a human does.

BEHAVIOR z-score · velocity · geo · time of day

LEARN XGBoost classifier spaces normal from fraud

CONNECT Neo4j graph finds fraud rings & shared devices

FUSE Ensemble scorer → combined risk 0–1

ACT Agentic triage builds a risk dossier

A real anomaly, told simply

Amit's account

usual spend ≈ $150 · always from IN · between 9am–9pm

Sudden $5,000 transfer

z□score 3.2 · velocity spike · geo jump to a new country

→ combined risk 0.78 → BLOCK + investigate

The agent gathers evidence into a dossier an analyst can trust in seconds — not hours.

Merchant approval — days down to minutes

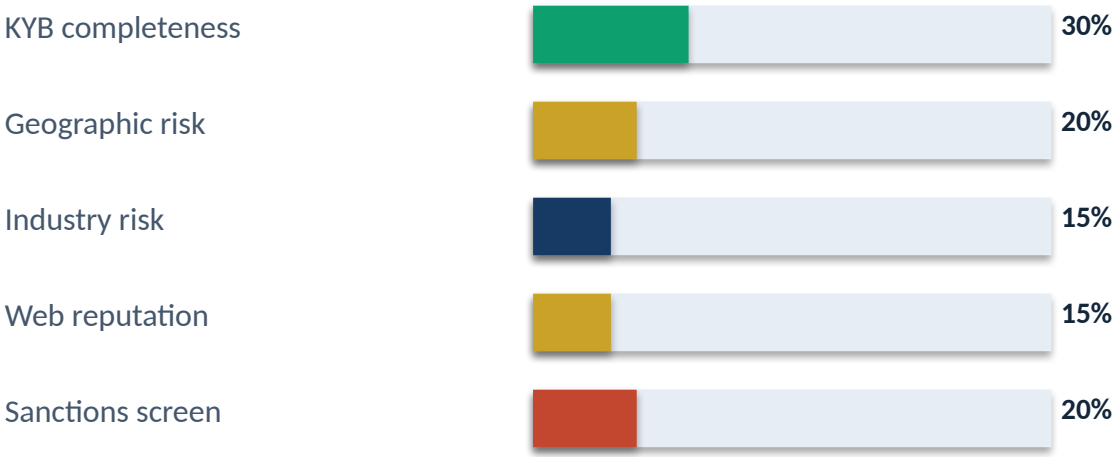
RAG document extraction + KYB verification + a transparent 5-signal risk score.

RAG + dual OCR parses registration docs (Tesseract speed + EasyOCR accuracy)

KYB verification checks required fields, industry & geographic risk against a high-risk country list

Result Acme Corp → 0.15 → APPROVE · Suspicious Ltd → 0.85 → REJECT

Transparent 5-signal risk score



The human layer

Every score is explained by its parts — so a rejected merchant knows why, and a reviewer can override with full context.

Four AI techniques, one explainable brain

Each capability is built so a regulator — and a customer — can follow the reasoning.

LLM Entity Resolution
Understands names, not strings

Fuzzy matching + LLMs recover aliases & transliterations →

98.2% match F1

Graph Neural Networks
Sees relationships

Neo4j graphs surface fraud rings & shared devices a rule can't →

97.5% detection

RAG for Compliance
Reads documents

Vector cloud of rules + OCR extracts facts with citations →

95% doc accuracy

Agentic AI Triage
Investigates first

An agent queries every source and returns a human risk dossier →

>85% analyst agreement

Measured results — tested, not promised

42 passing tests, 52% coverage, and targets met across every module.

0.98
Fraud detection AUC–ROC

100%
Screening recall (target >95%)

~8%
False-positive rate (target <10%)

96%
F1 score on fraud set

<5 min
Auto merchant onboarding (was days)

<100ms
p99 screening latency

Testing rigor 42 unit, integration & API tests across all modules · TDD throughout · Locust load tests

The human check triage decisions agree with human analysts >85% of the time — the machine assists, the analyst decides.

What changes for the people inside the system

Beyond metrics — real relief for analysts, merchants, and customers.

ANALYST	**from 100 alerts to the few that matter**	$50–$100 saved per false alarm; more time on real crime
MERCHANT	**onboarding in minutes, not days**	honest businesses reach customers before rivals do
CUSTOMER	**money moves safely, and stays explainable**	a blocked transfer comes with a reason, not a denial

Open-source & self-hosted 60–80% lower cost than commercial suites · explainable & auditable for the EU AI Act · deploy anywhere

A stack built to run, measure, and harden

Event-driven microservices on a containerized, Kubernetes-ready foundation.

RUNTIME

- FastAPI + Uvicorn
- Pydantic v2 · SQLAlchemy 2.0
- Docker + Docker Compose

DATA

- PostgreSQL 16
- Redis 7 · Kafka 7.6
- Milvus 2.4 · Neo4j 5.22

AI / ML

- LangChain · LlamaIndex
- sentence-transformers
- XGBoost · PyTorch Geometric

OPS

- Prometheus + Grafana
- GitHub Actions CI/CD
- Kubernetes (prod)

Reliability by design every module degrades gracefully · JWT auth + Redis rate limiting · structured audit = immutable JSON trail · health checks on all 10 services

“Built like a real product, not a class exercise” — one command spins up the whole stack; the code is open, readable, and ready for production evaluation.

From prototype to platform

A clear, honest path to production — and to wider, safer finance.

LIVE LLMs	Connect real GPT/Anthropic APIs for entity resolution & extraction
GRAPH AI	PyTorch Geometric GNNs for deeper fraud-ring detection
REGULATORY	Multi-jurisdiction rules · SAR auto-filing · PEP screening
SCALE	Kubernetes with HPA · multi-region · event replay
CRYPTO	On-chain analytics & mixer / VASP screening

Why we built it this way the architecture was designed so each future step lands without rewriting what works — compliance tech that can grow with the institutions that trust it.

Thank you

Questions & discussion

trust is the product — Sentinel just makes it measurable

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github.com/celestialgeeks/sentinel-fintech